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Change-Point Detection for Object-Valued Time Series

Yi Zhang^a, Changbo Zhu^b , and Xiaofeng Shao^c

^aDepartment of Statistics, University of Illinois at Urbana-Champaign, Champaign, IL; ^bDepartment of Applied and Computational Mathematics and Statistics, University of Notre Dame, Notre Dame, IN; ^cDepartment of Statistics and Data Science, and Department of Economics, Washington University in St. Louis, St. Louis, MO

ABSTRACT

This article is concerned with change point detection for object-valued data that reside in a metric space, which has attracted some recent interests in statistics and econometrics literature. The existing methods either focus on independent data or can only detect change in the Fréchet mean or variance. In this article, we propose a self-normalization (SN, hereafter) based statistic for detecting a shift in the marginal distribution of object-valued time series. Our test is universally applicable to a wide range of object-valued data, such as distributional and network data, and can accommodate weak serial dependence. In addition the proposed test statistic is almost tuning parameter free, has pivotal limiting null distribution and only uses the pairwise distances. When combined with the Wild Binary Segmentation algorithm (WBS, hereafter), our statistic can be used to estimate the number and locations of multiple change points. Asymptotic results for our SN based statistic are derived under both null and local alternatives in the single change point setting. For the first time, the WBS estimation consistency is shown for a broad class of object-valued time series and in a nonparametric setting, which requires new nonstandard theoretical arguments. Extensive numerical experiments and real data analysis are conducted to illustrate the effectiveness and broad applicability of our proposed method.

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1. Introduction

Statistical analysis of non-Euclidean data (object-valued data or random objects) that reside in a metric space is gradually emerging as an important branch of functional data analysis in econometrics and statistics, motivated by increasing encounter of such data in many modern applications. Examples include the analysis of sequences of age-at-death distributions over calendar years (Mazzucco and Scarpa 2015; Shang and Hyndman 2017), covariance matrices in the analysis of diffusion tensors in medical imaging (Dryden, Koloydenko, and Zhou 2009), compositional brain volumes (Zhu et al. 2023; Zhou et al. 2023) and graph Laplacians of networks (Ginestet et al. 2017; Wang, Yu, and Rinaldo 2021). One of the main challenges in dealing with such data is that the usual vector/Hilbert space operations, such as projection and inner product may not be well defined and only the distance between two non-Euclidean data objects is available. Despite the challenge, the list of papers that propose new statistical techniques to analyze non-Euclidean data has been growing. See Dubey and Müller (2019, 2020), Petersen and Müller (2019), Tucker, Wu, and Müller (2023), Zhou and Müller (2022), Zhang, Xue, and Li (2023), Zhu and Müller, (2023, 2024), Jiang, Gao, and Shao (2024a), Jiang, Zhu, and Shao (2024b), among others.

In this article, we shall focus on offline change-point detection for object-valued time series. Such temporally dependent

objects are prevalent in practice and well-known examples include yearly age-at-death distributions for countries in Europe (Dubey and Müller 2020), annual observations of compositions of energy sources for power generation in the United States (Zhu and Müller 2024), and daily Pearson correlation matrices for five cryptocurrencies (Jiang, Zhu, and Shao 2024b). Also see Section 5 for additional data examples. The literature for offline change-point detection of object-valued data is quite recent. For example, Chen and Zhang (2015) and Chu and Chen (2019) developed graph-based scan statistics for both the single change-point and the changed interval alternatives to detect the distributional change for independent object data; see Chen and Chu (2023) for a recent review on graph-based change point detection methods. Recently, Dubey and Müller (2020) proposed a novel test for a change point in the Fréchet mean and variance (Fréchet 1948) for a sequence of independent random objects and adopted binary segmentation for change point estimation. Dubey, Chen, and Müller (2022) proposed the concept of distance profiles, which fully characterize the distribution of random objects under conditions that basically require the metric space to be of strong negative type (see proposition 1 therein). Building on distance profile, Dubey and Zheng (2023) developed a test which can detect distributional change for serially independent random objects. Note that theoretical results developed for the above-mentioned methods

are all for single change point detection. To accommodate serial dependence, Jiang, Zhu, and Shao (2024b) developed an self-normalization (SN, hereafter) based test by extending the SN idea (Shao 2010; Shao and Zhang 2010; Shao 2015) from vector time series to object-valued time series. However, the method in Jiang, Zhu, and Shao (2024b) is also limited to capture change point in Fréchet mean or variance only and their test may not be powerful when the change is in high-order characteristics of distributions; see numerical evidence in Section 4.

In contrast to the limited literature for change-point detection of object-valued data, the same problem for functional time series, which resides in a Hilbert space, has been extensively studied. For example, single and multiple change point detection problem for functional data has been studied in Zhang et al. (2011), Aston and Kirch (2012), Sharipov, Tewes, and Wendler (2016), Aue, Rice, and Sönmez (2018), Jiao, Frostig, and Ombao (2023), Chiou, Chen, and Hsing (2019), Rice and Zhang (2022), Harris, Li, and Tucker (2022); see Chapter 8 in Horváth and Rice (2024) for a comprehensive review of the main developments in this area. It is worth noting that these methods can not be easily extended to the object-valued time series setting, due to the lack of algebraic operations for metric space valued data.

The main goal of this article is to fill in the gap and develop new testing and estimation procedures that target the shift in marginal distributions of object-valued time series with serial dependence. On the testing front, we propose a nonparametric method that can detect distributional changes for metric space valued time series based on sample splitting and self-normalization. To construct our test statistic, the object-valued data are first mapped into a Hilbert space, and then projected along the direction of change. After that, an SN-based test statistic is constructed using the projected one-dimensional sequence. In the single change point setting, the statistic we developed is nearly tuning parameter free (except for a cutoff parameter which restricts the true change point from being too close to the end-points) and has pivotal limiting null distribution. On the estimation front, we use the WBS algorithm (Fryzlewicz 2014) for multiple change point estimation and provide theoretical guarantee in terms of estimation consistency, which is a new contribution to the literature.

As one of the state-of-the-art segmentation algorithm with promising practical performance, WBS has been combined with various single change point detection methods for Euclidean or metric space valued time series to enable multiple change point estimation (see e.g., Wang and Samworth 2017; Chakraborty and Zhang 2021; Wang et al. 2022; Zhang, Wang, and Shao 2022; Jiang, Zhu, and Shao 2024b) without theoretical justification. The WBS algorithm was originally developed to detect change points in the mean of univariate iid data, and its performance on temporally dependent data could depend on the estimation of the long run variance (LRV), which was known to be a difficult problem due to the difficulty of choosing the bandwidth in the presence of change-points (Crainiceanu and Vogelsang 2007; Shao and Zhang 2010; Robbins et al. 2011). By using self-normalization, our method avoids direct estimation of LRV. At the same time, the adoption of a self-normalizer and the usage of projection in our test statistic significantly complicate our proof for estimation consistency, which requires substantially different theoretical arguments from those in Fryzlewicz

(2014). To our knowledge, except for Korkas and Fryzlewicz (2017) where the WBS estimation consistency was proved for a parametric time series model, we are the first to show the WBS estimation consistency for a broad class of time series (which includes Euclidean valued time series as a special case) and in a nonparametric setting. In our theory, the convergence rates for each change point estimator are adaptive to the heterogeneous change magnitudes (see Remark 5 in Section 3). A more detailed discussion on WBS and WBS2 (Fryzlewicz 2020) is provided at the beginning of Section 3.

The rest of this article is organized as follows. In Section 2, we first provide preliminary result on Functional CLT for Hilbert space valued time series in Section 2.1, and introduce our SN based method for single change point testing and estimation in Section 2.2. In Section 2.3, we provide some (counter) examples of metric spaces of strong negative type, and show that the metric spaces where some frequently encountered object-valued data live in are of strong negative type, which may be of independent interest. The WBS-SN algorithm for multiple change point estimation, as well as the estimation consistency result, are provided in Section 3. Section 4 demonstrates the promising performance of our proposed testing and estimation methods in finite sample via simulations. In Section 5, two real data applications are provided, including stock return data in Section 5.1 and TLC Trip Record Data in Section 5.2, respectively. Section 6 concludes and all the proofs are gathered in the supplement.

The following notations will be used throughout the article. Let $\|\cdot\|$ be the Euclidean norm on $\mathbb{R}^p$ and $\|\cdot\|_{\mathcal{H}} = \sqrt{\langle \cdot, \cdot \rangle_{\mathcal{H}}}$ be the norm on the Hilbert space $\mathcal{H}$ (all the Hilbert spaces considered in this article are separable (real) Hilbert spaces). For $\mathbb{R}$ - or $\mathcal{H}$ -valued random variables $\{X_t\}_{t \in \mathbb{Z}}$, let $\mathcal{F}_a^b$ denote the σ -field generated by $\{X_a, \dots, X_b\}$ for any $-\infty \leq a \leq b \leq \infty$. We use “ $\rightsquigarrow$ ” to denote the weak convergence in $D_{\mathcal{H}}[0, 1]$: the space of $\mathcal{H}$ -valued cadlag functions defined on $[0, 1]$ endowed with the Skorokhod metric.

2. Single Change Point Estimation

2.1. Preliminary Result

Given $\mathbb{R}$ - or $\mathcal{H}$ -valued random variables $\{X_t\}_{t \in \mathbb{Z}}$ and positive integer n , define its ρ -mixing coefficient (Bradley 2005) as $\rho(n) = \sup_{j \in \mathbb{Z}} \rho(\mathcal{F}_{-\infty}^j, \mathcal{F}_{j+n}^{\infty})$, where

$$\rho(\mathcal{A}, \mathcal{B}) = \sup \left\{ \frac{|\text{cov}(X, Y)|}{\sqrt{\text{var}(X) \text{var}(Y)}} : X \in L_2(\mathcal{A}), Y \in L_2(\mathcal{B}) \right\}$$

and $L_2(\mathcal{A})$ is the set of $\mathbb{R}$ -valued $\mathcal{A}$ -measurable random variables with finite second moment for any σ -algebra $\mathcal{A}$. $\{X_t\}_{t \in \mathbb{Z}}$ is said to be ρ -mixing if $\lim_{n \rightarrow \infty} \rho(n) = 0$. As in Buccia and Wendler (2017), define the coefficient $\alpha_{1,1}(n) = \sup_{j \in \mathbb{Z}, k \geq n} \alpha(\mathcal{F}_j^j, \mathcal{F}_{j+k}^{j+k})$, where $\alpha(\mathcal{A}, \mathcal{B}) = \sup \{|P(A \cap B) - P(A)P(B)| : A \in \mathcal{A}, B \in \mathcal{B}\}$ for any σ -fields $\mathcal{A}$ and $\mathcal{B}$. An operator $Q : \mathcal{H} \rightarrow \mathcal{H}$ is said to be a positive, self-adjoint, trace class operator if (i) $\langle Qh, h \rangle_{\mathcal{H}} \geq 0$ for any $h \in \mathcal{H}$; (ii) $\langle Qh, x \rangle_{\mathcal{H}} = \langle h, Qx \rangle_{\mathcal{H}}$ for any $h, x \in \mathcal{H}$; (iii) $\sum_{k=1}^{\infty} \langle Qe_k, e_k \rangle_{\mathcal{H}} < \infty$ for any orthonormal base $\{e_k\}$ of $\mathcal{H}$.

Based on these concepts, we can define $\mathcal{H}$ -valued Gaussian random element as well as $\mathcal{H}$ -valued Brownian motion.

Definition 1 (Andresen, Imkeller, and Perkowski (2013)). An $\mathcal{H}$ -valued random element X is Gaussian with covariance operator Q and mean $\mu \in H$ (denoted as $X \sim \mathcal{N}(\mu, Q)$) if for any $h \in \mathcal{H}$, $\langle X, h \rangle_{\mathcal{H}} \sim \mathcal{N}(\langle \mu, h \rangle_{\mathcal{H}}, \langle Qh, h \rangle_{\mathcal{H}})$.

Definition 2 (Andresen, Imkeller, and Perkowski (2013)). Let Q be a positive, self-adjoint, trace class operator on $\mathcal{H}$. An $\mathcal{H}$ -valued Brownian motion with covariance operator Q is defined as $B_Q(t), t \in [0, 1]$ such that (i) $B_Q(0) = 0$; (ii) $B_Q(t)$ has continuous sample path; (iii) $B_Q(t)$ has independent increment; (iv) $B_Q(t) - B_Q(s) \sim \mathcal{N}(0, (t-s)Q)$ for any $0 \leq s < t \leq 1$.

The following FCLT for stationary $\mathcal{H}$ -valued random variables is from Bucchia and Wendler (2017); see Sharipov, Tewes, and Wendler (2016) for similar results under different mixing conditions.

Theorem 1. Let $\{X_t\}_{t \in \mathbb{Z}}$ be a sequence of strictly stationary $\mathcal{H}$ -valued random variables with mean μ . Assume $\{X_t\}_{t \in \mathbb{Z}}$ is ρ -mixing and the following condition holds for some $\delta > 0$: (1) $\mathbb{E} \|X_1\|_{\mathcal{H}}^{2+\delta} < \infty$; (2) $\sum_{n=1}^{\infty} [\alpha_{1,1}(n)]^{\delta/(2+\delta)} < \infty$. Then $\frac{1}{\sqrt{n}} \sum_{t=1}^{\lfloor nr \rfloor} (X_t - \mu) \rightsquigarrow B_Q(t)$, where $\{B_Q(t)\}_{t \in [0,1]}$ is an $\mathcal{H}$ -valued Brownian motion with

$$\langle Qx, y \rangle_{\mathcal{H}} = \sum_{t \in \mathbb{Z}} \mathbb{E} (\langle X_0 - \mu, x \rangle_{\mathcal{H}} \langle X_t - \mu, y \rangle_{\mathcal{H}}) \text{ for any } x, y \in \mathcal{H}. \quad (1)$$

Furthermore, the series in (1) converges absolutely.

2.2. Single Change Point Testing and Estimation

From now on, we assume the time series $\{X_t\}_{t \in \mathbb{Z}}$ takes value in a separable metric space $(\mathcal{X}, d)$ of strong negative type (see Definition 2.2 in Chakraborty and Zhang (2021) or Lyons (2013)). Let P_i be the marginal distribution of X_i . We aim to test $H_0 : P_1 = \dots = P_n$ versus $H_1 : P_1 = \dots = P_{k^*} \neq P_{k^*+1} = \dots = P_n$ for some $k^* = \lfloor nr_0 \rfloor \in (nb, (1-b)n)$ with $b \in (0, 1/2)$. Here b is a cutoff parameter and it is typically set to be 0.1 or 0.15; see Andrews (1993). This is a common assumption as the detection of a change point is very challenging when it is close to the boundary; see Horvath, Miller, and Rice (2020) for some recent work.

By Proposition 3.1 in Chakraborty and Zhang (2021), for a metric space $(\mathcal{X}, d)$ of strong negative type, there exists a Hilbert space $\mathcal{H}$ and an embedding map $\phi : \mathcal{X} \rightarrow \mathcal{H}$ such that $d(x, x') = \|\phi(x) - \phi(x')\|_{\mathcal{H}}^2$. For $t \in \mathbb{Z}$, let $Y_t = \phi(X_t)$. If $\int \|Y_t\|_{\mathcal{H}} dP_t < \infty$, define $\mu_t = \int Y_t dP_t$, which is the unique element in $\mathcal{H}$ corresponding to the bounded linear functional $\mathcal{H} \rightarrow \mathbb{R} : h \rightarrow \int \langle h, Y_t \rangle_{\mathcal{H}} dP_t$. Then by Proposition 3.1 in Lyons (2013), the null and alternative hypothesis are equivalent to $H_0 : \mu_1 = \dots = \mu_n$ versus $H_1 : \mu_1 = \dots = \mu_{k^*} \neq \mu_{k^*+1} = \dots = \mu_n$.

Remark 1. Some possible embedding maps for specific metric spaces are given in sec. 3 of Lyons (2013). For example, when $(\mathcal{X}, d)$ is the one dimensional Euclidean space, we can define $\phi(x) = \mathbb{1}_{[0, \infty)}(y) - \mathbb{1}_{[x, \infty)}(y) \in L^2(\mathbb{R})$ where $L^2(\mathbb{R})$ is the Hilbert space of square integrable functions on $\mathbb{R}$. A similar embedding into $L^2(\mathbb{R}^p)$ can also be defined when $(\mathcal{X}, d)$ is the

p -dimensional Euclidean space for $p \geq 2$. Additionally, define $k : \mathbb{R}^p \times \mathbb{R}^p \rightarrow \mathbb{R}$ as

$$k(x, y) = \frac{1}{2} [d(x, x_0) + d(y, x_0) - d(x, y)] \quad (2)$$

for some fixed $x_0 \in \mathbb{R}^p$. Then we can also define the embedding of the p -dimensional Euclidean space as $\phi(x) = k(x, \cdot)$ where $k(x, \cdot)$ is an element in the reproducing kernel Hilbert space (RKHS) induced by the kernel function $k(\cdot, \cdot)$; see sec. 4 of Sejdinovic et al. (2013) for more details.

To detect a change in μ_t , we first split the data into three parts: $\mathcal{X}_1 = \{X_1, \dots, X_{\lfloor nb \rfloor}\}$, $\mathcal{X}_2 = \{X_{\lfloor nb \rfloor+1}, \dots, X_{n-\lfloor nb \rfloor}\}$ and $\mathcal{X}_3 = \{X_{n-\lfloor nb \rfloor+1}, \dots, X_n\}$. Based on $\mathcal{X}_1$ and $\mathcal{X}_3$, we define

$$\bar{Y}_1 - \bar{Y}_3 = \frac{1}{\sqrt{n}} \sum_{t=1}^{\lfloor nb \rfloor} Y_t - \frac{1}{\sqrt{n}} \sum_{t=n-\lfloor nb \rfloor+1}^n Y_t, \quad (3)$$

which estimates the mean change $\mu_1 - \mu_n$ (subject to a rescaling factor) under H_1 . Then we project the transformed observations in $\mathcal{X}_2$ along the direction $\bar{Y}_1 - \bar{Y}_3$, that is, we define univariate random variables $Z_t = \langle \bar{Y}_1 - \bar{Y}_3, Y_t \rangle_{\mathcal{H}}$ for $t = \lfloor nb \rfloor+1, \dots, n-\lfloor nb \rfloor$. Note that the temporal dependence in the projected sequence $\{Z_t\}$ is very complex due to projection. Nevertheless, we shall apply the SN test statistic in Shao and Zhang (2010), which is developed for weakly dependent vector time series, to $\{Z_t\}$. To be specific, let $S_{a,b} = \sum_{t=a}^b Z_t$, our test statistic is defined as

$$G_n = \max_{k=\lfloor nb \rfloor+1, \lfloor nb \rfloor+2, \dots, n-\lfloor nb \rfloor-1} \frac{T_n(k)}{\sqrt{V_n(k)}},$$

where T_n, V_n are defined as

$$T_n(k) = \frac{1}{\sqrt{n-2\lfloor nb \rfloor}} \sum_{t=\lfloor nb \rfloor+1}^k \left(Z_t - \frac{S_{\lfloor nb \rfloor+1, n-\lfloor nb \rfloor}}{n-2\lfloor nb \rfloor} \right), \quad (4)$$

$$V_n(k) = \frac{1}{(n-2\lfloor nb \rfloor)^2} \left\{ \sum_{t=\lfloor nb \rfloor+1}^k \left\{ S_{\lfloor nb \rfloor+1, t} - \frac{t-\lfloor nb \rfloor}{k-\lfloor nb \rfloor} S_{\lfloor nb \rfloor+1, k} \right\}^2 + \sum_{t=k+1}^{n-\lfloor nb \rfloor} \left\{ S_{t, n-\lfloor nb \rfloor} - \frac{n-\lfloor nb \rfloor-t+1}{n-\lfloor nb \rfloor-k} S_{k+1, n-\lfloor nb \rfloor} \right\}^2 \right\}. \quad (5)$$

Remark 2. It is worth noting that the sequence $\{Y_t\}$ depends on the embedding map $\phi(\cdot)$ which may be unknown or difficult to derive. However, due to the use of sample splitting and self-normalization, our statistic G_n does not depend on $\phi(\cdot)$ in any explicit way and only depends on pairwise distances. To see this, note that

$$\begin{aligned} Z_t &= \frac{1}{\sqrt{n}} \sum_{j=1}^{\lfloor nb \rfloor} \langle Y_j, Y_t \rangle_{\mathcal{H}} - \frac{1}{\sqrt{n}} \sum_{j=n-\lfloor nb \rfloor+1}^n \langle Y_j, Y_t \rangle_{\mathcal{H}} \\ &= \frac{1}{2\sqrt{n}} \sum_{j=1}^{\lfloor nb \rfloor} \{ \|Y_j\|_{\mathcal{H}}^2 + \|Y_t\|_{\mathcal{H}}^2 - d(X_j, X_t) \} \end{aligned}$$

$$\begin{aligned}
& -\frac{1}{2\sqrt{n}} \sum_{j=n-\lfloor nb \rfloor+1}^n \{ \|Y_j\|_{\mathcal{H}}^2 + \|Y_t\|_{\mathcal{H}}^2 - d(X_j, X_t) \} \\
& = -\frac{1}{2\sqrt{n}} \sum_{j=1}^{\lfloor nb \rfloor} d(X_j, X_t) + \frac{1}{2\sqrt{n}} \sum_{j=n-\lfloor nb \rfloor+1}^n d(X_j, X_t) + R_n,
\end{aligned}$$

where $R_n = \frac{1}{2\sqrt{n}} \sum_{j=1}^{\lfloor nb \rfloor} \|Y_j\|_{\mathcal{H}}^2 + \frac{1}{2\sqrt{n}} \sum_{j=n-\lfloor nb \rfloor+1}^n \|Y_j\|_{\mathcal{H}}^2$ is an unknown quantity that depends on $\phi(\cdot)$. The key is that R_n does not depend on t , so it gets canceled out when forming the SN test statistic. To be specific, define $\hat{Z}_t = \sum_{j=n-\lfloor nb \rfloor+1}^n d(X_j, X_t) - \sum_{j=1}^{\lfloor nb \rfloor} d(X_j, X_t)$ and $\hat{S}_{a,b} = \sum_{t=a}^b \hat{Z}_t$, then the statistic G_n can be calculated by replacing $Z_t, S_{s,b}$ in the definition of $T_n(k)$ and $V_n(k)$ with $\hat{Z}_t$ and $\hat{S}_{a,b}$. In other words, our test statistic G_n only depends on the pairwise distances $\{d(X_t, X_{t'})\}_{t,t'=1}^n$.

Remark 3. Note that our statistic G_n only requires one trimming parameter b , whereas the SN-based statistics proposed in sec. 4 of Jiang, Zhu, and Shao (2024b) require two tuning parameters η_1 and η_2 , and target the changes in Fréchet mean and variance only. In their paper, η_1 plays the same role as our trimming parameter b and η_2 ensures their subsample estimators of Fréchet mean and variance are calculated with large enough subsample size. Since the statistics in Jiang, Zhu, and Shao (2024b) require repeated estimation of Fréchet mean and variance on different subsamples, they are much more expensive than G_n in computation.

As shown below, the limiting null distribution of G_n is G , which is pivotal and its upper critical values have been tabulated in Gao, Wang, and Shao (2023). Given the level $\gamma \in (0, 1)$ (say, 0.05), our corresponding test is $\mathbf{1}(G_n > G_\gamma)$, where G_γ is the $100(1 - \gamma)$ th upper percentile of G . The following assumptions are needed to derive the asymptotic distributions of G_n under the null and alternative.

Assumption 1. Let $\{Y_t - \mu_t\}_{t \in \mathbb{Z}}$ be a strictly stationary $\mathcal{H}$ -valued time series. Assume $\{Y_t\}_{t \in \mathbb{Z}}$ is ρ -mixing and the following condition hold for some $\delta > 0$:

1. $\mathbb{E} d(X_1, x_0)^{1+\delta/2} < \infty$ for some $x_0 \in \mathcal{X}$;
2. $\sum_{n=1}^{\infty} [\alpha_{1,1}(n)]^{\delta/(2+\delta)} < \infty$.

Remark 4. By the definition of $\phi(\cdot)$, we can see it is a continuous bijection between $(\mathcal{X}, d)$ and $(\phi(\mathcal{X}), \|\cdot\|_{\mathcal{H}})$ with continuous inverse. For any positive integers $a \leq b$, the σ -algebra generated by $\{Y_a, \dots, Y_b\}$ is the same as the σ -algebra generated by $\{X_a, \dots, X_b\}$. So part 2 in Assumption 1 holds for $\{Y_t\}$ if and only if it holds for $\{X_t\}$.

By replacing $\phi(\cdot)$ with $\phi(\cdot) - \phi(x_0)$, we can assume, without loss of generality, that $\phi(x_0) = 0 \in \mathcal{H}$. Then $\mathbb{E} \|Y_1\|_{\mathcal{H}}^{2+\delta} = \mathbb{E} d(X_1, x_0)^{1+\delta/2}$. Note that

$$\begin{aligned}
& \mathbb{E} \langle Y_t - \mu_t, Y_t - \mu_t \rangle_{\mathcal{H}}^{1+\delta/2} \\
& = \mathbb{E} \langle Y_1 - \mu_1, Y_1 - \mu_1 \rangle_{\mathcal{H}}^{1+\delta/2} \\
& \leq 2^{1+\delta} (\mathbb{E} \|Y_1\|_{\mathcal{H}}^{2+\delta} + \|\mu_1\|_{\mathcal{H}}^{2+\delta}) \\
& = 2^{1+\delta} \mathbb{E} d(X_1, x_0)^{1+\delta/2}
\end{aligned} \tag{6}$$

$$\begin{aligned}
& + 2^{1+\delta} \left[\int \int \langle \phi(x), \phi(y) \rangle_{\mathcal{H}} dP_1(x) dP_1(y) \right]^{1+\delta/2} \\
& \leq 2^{2+\delta} \mathbb{E} d(X_1, x_0)^{1+\delta/2},
\end{aligned}$$

so part 1 in Assumption 1 implies that $\mathbb{E} \|Y_t - \mu_t\|_{\mathcal{H}}^{2+\delta} < \infty$, which further implies the existence of μ_t . By Theorem 1 we have $\sum_{t=1}^{\lfloor nr \rfloor} (Y_t - \mu_t) / \sqrt{n} \rightsquigarrow B_Q(t)$, where $\{B_Q(t)\}_{t \in [0,1]}$ is an $\mathcal{H}$ -valued Brownian motion with $\langle Qx, y \rangle_{\mathcal{H}} = \sum_{t \in \mathbb{Z}} \mathbb{E} (\langle Y_0 - \mu_0, x \rangle_{\mathcal{H}} \langle Y_t - \mu_t, y \rangle_{\mathcal{H}})$ for any $x, y \in \mathcal{H}$. The following theorem shows the asymptotic properties of G_n , which is proved in the appendix of the supplement.

Theorem 2. Suppose Assumption 1 holds, then: (i) under H_0 , we have $G_n \xrightarrow{\mathcal{D}} G$, where

$$G \stackrel{d}{=} \sup_{r \in [0,1]} \frac{B(r) - rB(1)}{\left\{ \int_0^r [B(s) - \frac{s}{r}B(r)]^2 ds + \int_r^1 [B(1) - B(s) - \frac{1-s}{1-r}(B(1) - B(r))]^2 ds \right\}^{1/2}},$$

$\{B(t)\}_{t \in [0,1]}$ is an $\mathbb{R}$ -valued standard Brownian motion; (ii) under H_A , denote $\Delta_n = \mu_n - \mu_1$, we have

1. If $\sqrt{n} \|\Delta_n\|_{\mathcal{H}} \rightarrow \infty$, then $G_n \xrightarrow{P} \infty$.
2. If $\sqrt{n} \Delta_n \rightarrow c \in \mathcal{H}$ and $\|c\|_{\mathcal{H}} \neq 0$, then we have

$$\bar{Y}_1 - \bar{Y}_3 \xrightarrow{\mathcal{D}} B_Q(b) - [B_Q(1) - B_Q(1-b)] - bc = Y^*, G_n \xrightarrow{\mathcal{D}} G^*,$$

where the conditional distribution of G^* given $Y^* = y$ is equal to the distribution of

$$\sup_{r \in [0,1]} \frac{B'(r) - rB'(1)}{\left\{ \int_0^r [B'(s) - \frac{s}{r}B'(r)]^2 ds + \int_r^1 [B'(1) - B'(s) - \frac{1-s}{1-r}(B'(1) - B'(r))]^2 ds \right\}^{1/2}},$$

$B'(r) = B(r) + H_y((1-2b)r+b)/\sqrt{1-2b}$, $H_y = \langle y, (r-r_0)c \mathbf{1}_{r \geq r_0} \rangle_{\mathcal{H}} / \sqrt{\langle Qy, y \rangle_{\mathcal{H}}}$ with $\mathbf{1}_{r \geq r_0} = 1$ if $r \geq r_0$ and 0 otherwise.

3. If $\sqrt{n} \|\Delta_n\|_{\mathcal{H}} \rightarrow 0$, then $G_n \xrightarrow{\mathcal{D}} G$, so our test have trivial power asymptotically.

When a single change point is detected, it is of interest to estimate its location. Building upon the test statistic G_n , it is natural to estimate k^* by

$$\hat{k} = \arg \max_{k=\lfloor nb \rfloor+1, \lfloor nb \rfloor+2, \dots, n-\lfloor nb \rfloor-1} \frac{T_n(k)}{\sqrt{V_n(k)}}.$$

The following theorem shows the consistency of $\hat{k}$, which is proved in the appendix of the supplement.

Theorem 3. Suppose Assumption 1 holds and $\sqrt{n} \|\Delta_n\|_{\mathcal{H}} \rightarrow \infty$. For a given integer sequence ϵ_n which satisfies that $\frac{\epsilon_n}{n} \rightarrow 0$ and $n(\epsilon_n \|\Delta_n\|_{\mathcal{H}})^{-2} \rightarrow 0$, we have

$$P(|\hat{k} - k^*| < \epsilon_n) \rightarrow 1. \tag{7}$$

The proof of Theorem 3 basically follows the same idea as in the proof of Theorem 1 in Zhao, Jiang, and Shao (2022). Note that the convergence rate ϵ_n of the change point estimator $\hat{k}$ is the same as the rate derived in Theorem 1 of Zhao, Jiang, and Shao (2022), which first worked out the consistency and the rate

for the SN-based change point location estimator for Euclidean time series. The latter paper also stated that under the fixed alternative setting (i.e., $\|\Delta_n\|_{\mathcal{H}} = C > 0$), the convergence rate ϵ_n/n is at best $1/\sqrt{n}$. In comparison, as shown in Bai (1994), the optimal convergence rate for detecting change point in the mean of a multivariate time series is $1/n$. It seems difficult to improve convergence rate ϵ_n due to the complex form of $V_n(k)$ in the definition of G_n . We refer to the discussion following Theorem 4 in Zhao, Jiang, and Shao (2022) for more details of a local refinement procedure to achieve the optimal rate.

2.3. Strong Negativity of Some Metric Spaces

According to Theorem 3.16 in Lyons (2013), every separable Hilbert space, such as the Euclidean space or the space of square integrable functions, is a metric space of strong negative type. Besides that, it is in general not easy to verify that a metric space is of strong negative type. According to part (iv) in Lyons (2021), if $(\mathcal{X}, d)$ is a separable semimetric space of negative type, then $(\mathcal{X}, d^r)$ is a metric space of strong negative type for any $r \in (0, 1)$. So in order to show $(\mathcal{X}, d)$ is a metric space of strong negative type, it suffices to show, by verifying the definition, $(\mathcal{X}, d^r)$ is of negative type for some $r > 1$. For example, by Theorem 3.6 in Meckes (2013), it is immediately clear that the space of measurable functions supported on $[0, 1]$ with metric $d(f, g) = \{\int_{[0,1]} |f(x) - g(x)|^p dx\}^{1/p}$ is a metric space of strong negative type for any $p \in (1, 2)$.

As for metric spaces that are not of strong negative type, it is proved in Dor (1976) that $\mathbb{R}^q$ with the ℓ^p metric is not a metric space of negative type for any $3 \leq q \leq \infty$ and $2 < p \leq \infty$. In addition, as stated in Lyons (2013), $\mathbb{R}^2$ with the ℓ^1 metric is a metric space of negative type, but not of strong negative type.

We now show the following metric spaces where object-valued data resides in are of strong negative type.

1. The space Ω_1 of probability measures on $\mathbb{R}$, equipped with the 2-Wasserstein metric $d_W(x, y) = \{\int_0^1 (F_x^{-1}(u) - F_y^{-1}(u))^2 du\}^{1/2}$, where F_x^{-1}, F_y^{-1} are the inverse cdf functions of x and y .
2. The space Ω_2 of Graph Laplacians of weighted graphs (see sec. 3 in Ginestet et al. 2017) equipped with Frobenius metric $d_F(A, B) = \{\text{tr}[(A - B)^\top (A - B)]\}^{1/2}$ for $d \times d$ symmetric matrices A and B .
3. The space Ω_3 of positive definite covariance matrices, equipped with the log-Euclidean metric $d_E(A, B) = d_F(\log_m(A), \log_m(B))$ for $d \times d$ symmetric matrices A and B , where $\log_m$ is the matrix-log function.

According to part (iv) in Lyons (2021), it suffice to show (Ω_1, d_W^2) , (Ω_2, d_F^2) and (Ω_3, d_E^2) are separable semimetric spaces of negative type. The separability of (Ω_1, d_W^2) is implied by the separability of (Ω_1, d_W) , which is shown in Bolley (2008). For any $n \geq 2$, $x_1, x_2, \dots, x_n \in \Omega_1$, $a_1, a_2, \dots, a_n \in \mathbb{R}$ with $\sum_{i=1}^n a_i = 0$,

$$\begin{aligned} & \sum_{i=1}^n \sum_{j=1}^n a_i a_j d_W^2(x_i, x_j) \\ &= \int_0^1 \sum_{i=1}^n \sum_{j=1}^n a_i a_j (F_{x_i}^{-1}(u) - F_{x_j}^{-1}(u))^2 du \end{aligned}$$

$$\begin{aligned} &= -2 \int_0^1 \sum_{i=1}^n \sum_{j=1}^n a_i a_j F_{x_i}^{-1}(u) F_{x_j}^{-1}(u) du \\ &= -2 \int_0^1 \left(\sum_{i=1}^n a_i F_{x_i}^{-1}(u) \right)^2 du \leq 0, \end{aligned} \quad (8)$$

so (Ω_1, d_W^2) is of negative type and the strong negativity of (Ω_1, d_W) is shown. For (Ω_2, d_F^2) , the separability follows from the fact that $\mathbb{R}^{d(d+1)/2}$ with the Euclidean metric is separable and every subspace of a separable metric space is separable (see Proposition 26 in sec. 9.6 of Royden and Fitzpatrick 2010). Denote A_{rs} as the (r, s) element of a matrix $A \in \Omega_2$, for any $n \geq 2$, $A^1, A^2, \dots, A^n \in \Omega_2$, $a_1, a_2, \dots, a_n \in \mathbb{R}$ with $\sum_{i=1}^n a_i = 0$, we have

$$\begin{aligned} \sum_{i=1}^n \sum_{j=1}^n a_i a_j d_F^2(A^i, A^j) &= \sum_{r=1}^n \sum_{s=1}^n \sum_{i=1}^n \sum_{j=1}^n a_i a_j (A_{rs}^i - A_{rs}^j)^2 \\ &= -2 \sum_{r=1}^n \sum_{s=1}^n \left(\sum_{i=1}^n a_i A_{rs}^i \right)^2 \leq 0, \end{aligned}$$

so (Ω_2, d_F^2) is of negative type, which implies (Ω_2, d_F) is of strong negative type. The strong negativity of (Ω_3, d_E) can be shown in a similar way as (Ω_2, d_F) and we omit the details.

3. Multiple Change Point Estimation: The WBS-SN Algorithm

The original WBS algorithm was proposed in Fryzlewicz (2014) to detect changes in the mean f_t for $\mathbb{R}$ -valued random variables $X_t = f_t + \epsilon_t$ with noise $\epsilon_t \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$. It operates in an iterative manner as follows: At the beginning, M independent random intervals $\{(s_m, e_m)\}_{m=1}^M$ are drawn from $\{1, 2, \dots, n\}$ and $C_m = \max_{k \in \{s_m, \dots, e_m-1\}} |\tilde{T}_m(k)|$ are calculated where the CUSUM statistic $\tilde{T}_m(k)$ (calculated on $\{X_{s_m}, \dots, X_{e_m}\}$) is a rescaled version of $T_n(k)$ defined in (4). If $\max\{C_m : m = 1, \dots, M\} > K_n$ for some threshold K_n , then the first estimated change point is chosen as k_1 where $(m_1, k_1) = \arg \max_{m \in \{1, 2, \dots, M\}, k \in \{s_m, \dots, e_m-1\}} |\tilde{T}_m(k)|$. The same procedure is then repeated for C_m calculated on the intervals inside subsamples $\{X_1, \dots, X_{k_1}\}$ and $\{X_{k_1+1}, \dots, X_n\}$, which gives the second and third change point estimator. The splitting and testing continue until no change points are found in each subsample. That is, the algorithm stops if, after several iterations, the maximum of C_m calculated in all resulting subsamples are less than K_n .

As stated in sec. 3.4 of Fryzlewicz (2014), the performance of WBS is sensitive to the choice of the threshold K_n , which depends on the estimation of the noise level σ^2 . Alternatively, Fryzlewicz (2014) proposed another implementation of WBS by setting $K_n = 0$ in the above algorithm and select the change point estimators from the output of WBS using “strengthened Schwarz Information Criterion” (sSIC), which also depends on an estimator of σ^2 . The extension of WBS to mean change point estimation in time series data inevitably involves the LRV estimation, which is a difficult task due to the bandwidth selection and the presence of change points. The popularly adopted kernel estimator of LRV tends to incur downward bias (Chan and

Yau 2017) and requires careful choice of bandwidth parameters. So far there seems very little theoretical work on the consistency of WBS in the time series setting. To the best of our knowledge, the only consistency result for the use of WBS was presented in Korkas and Fryzlewicz (2017), which is developed for detecting variance change in a sequence of random scaled χ_1^2 -distributed time series data with piecewise constant variance, where the CUSUM statistic $\tilde{T}_m(k)$ is scaled by $q_m = (e_m - s_m + 1)^{-1} \sum_{t=s_m}^{e_m} X_t^2$.

Recently, Fryzlewicz (2020) proposed the Wild Binary Segmentation 2 (WBS2) algorithm for iid data, where instead of generating M random intervals at the beginning, a smaller number of random intervals are drawn in each iteration and the algorithm stops after $(n-1)$ change point candidates are selected (see Theorem 3.1 in Fryzlewicz 2020). Based on WBS2, Cho and Fryzlewicz (2023) proposed the WCM.gSa algorithm for detecting multiple change points in the mean of time series data, where a new application of the Schwarz criterion (Schwarz 1978) is used to select the change point estimators from selected sub-collections in the output of WBS2. Although WBS2 does not impose any model assumption on the data, the estimation consistency result of WCM.gSa requires the demeaned time series coming from an AR(p) model.

In this section, we present an algorithm (WBS-SN) that estimates the locations of unknown but fixed number of change points in $\{\mu_t\}$ based on WBS. For positive integer m_0 and $r_0^* = 0 < r_1^* < r_2^* < \dots < r_{m_0}^* < 1 = r_{m_0+1}^*$, we assume the change points locate at $k_i^* = \lfloor nr_i^* \rfloor$ and have magnitude $\Delta_i^* = \mu_{k_i^*+1} - \mu_{k_i^*}$ for $i = 1, 2, \dots, m_0$. Then the data generating process under the multiple change alternative corresponds to

$$H_1 : P_1 = \dots = P_{k_1^*} \neq P_{k_1^*+1} = \dots = P_{k_2^*} \neq P_{k_2^*+1} \dots P_{k_{m_0}^*} \neq P_{k_{m_0}^*+1} = \dots = P_n.$$

We generate M small intervals such that the length of each interval is proportional to n in the following way. Let $\{U_{1m}\}_{m=1}^M, \{U_{2m}\}_{m=1}^M$ be two independent sequences of iid Uniform[0, 1] random variables, $\underline{U}_m = \min\{U_{1m}, U_{2m}\}$ and $\bar{U}_m = \max\{U_{1m}, U_{2m}\}$, then each interval is represented as $(\underline{U}_m, \bar{U}_m)$. Consider the event that each change point is

contained in the middle of at least one interval, which does not contain any other change points. To be specific, define

$$\mathcal{M} = \left\{ \text{For each } i = 1, 2, \dots, m_0, \text{ there exist } m \in \{1, 2, \dots, M\}, \text{ such that } r_{i-1}^* < \underline{U}_m < r_i^* < \bar{U}_m < r_{i+1}^* \text{ and } \frac{r_i^* - \underline{U}_m}{\bar{U}_m - \underline{U}_m} \in (b, 1 - b) \right\}. \quad (9)$$

Denote $\underline{r} = \min_{1 \leq i \leq m_0+1} (r_i^* - r_{i-1}^*) \leq \frac{1}{m_0+1}$, then we have

$$\begin{aligned} 1 - P(\mathcal{M}) &\leq \sum_{i=1}^{m_0} \left\{ 1 - P(\{r_{i-1}^* < \underline{U}_1 < r_i^* < \bar{U}_1 < r_{i+1}^* \text{ and } \frac{r_i^* - \underline{U}_1}{\bar{U}_1 - \underline{U}_1} \in (b, 1 - b)\}) \right\}^M \\ &\leq m_0 \left\{ 1 - \frac{2\underline{r}(1-2b)}{1-b} \right\}^M, \end{aligned}$$

which implies $P(\mathcal{M}) \rightarrow 1$ as $M \rightarrow \infty$. Denote $b_m = \lfloor (\lfloor n\bar{U}_m \rfloor - \lfloor n\underline{U}_m \rfloor + 1)b \rfloor$, and let $T_{nm}(k)$ and $V_{nm}(k)$ be the numerator and squared denominator of the SS-SN statistic, as defined in (4) and (5), which are calculated using data $\{X_{\lfloor n\underline{U}_m \rfloor}, X_{\lfloor n\underline{U}_m \rfloor+1}, \dots, X_{\lfloor n\bar{U}_m \rfloor}\}$ for some $k \in \{\lfloor n\underline{U}_m \rfloor + b_m, \dots, \lfloor n\bar{U}_m \rfloor - b_m - 1\}$. Using pseudocode, the WBS-SN algorithm is defined in Algorithm 1.

To start, we apply Algorithm 1 with WBS-SN(1, n, K_n, L_0, M, b). Here L_0 denotes the minimal interval length, b is the proportion of samples used to calculate the direction of projection $\bar{Y}_1 - \bar{Y}_3$ on each random small interval and K_n is the threshold. In practice, following the WBS procedure in Wang et al. (2022), K_n is determined by simulations as follows: we generate R samples of iid standard normal random variables with the same sample length n as the original object-valued data. For the r th sample, we calculate

$$\hat{K}_n^r = \max_{m \in \mathcal{M}_{1:n}, k \in \{\lfloor n\underline{U}_m \rfloor + b_m, \dots, \lfloor n\bar{U}_m \rfloor - b_m - 1\}} \frac{T_{nm}^{(r)}(k)}{\sqrt{V_{nm}^{(r)}(k)}}, \quad r = 1, 2, \dots, R, \quad (10)$$

Algorithm 1 WBS-SN for object-valued times series change point estimation

Function WBS-SN (s, e, K_n, L_0, M, b) :

```

if  $e - s < L_0$  then
  | STOP
else
   $\mathcal{M}_{s,e} :=$  set of those indices  $m \in \{1, 2, \dots, M\}$  for which  $[\lfloor n\underline{U}_m \rfloor, \lfloor n\bar{U}_m \rfloor] \subset [s, e]$  and  $\lfloor n\bar{U}_m \rfloor - \lfloor n\underline{U}_m \rfloor \geq L_0$ .
   $(m', k') := \arg \max_{m \in \mathcal{M}_{s,e}, k \in \{\lfloor n\underline{U}_m \rfloor + b_m, \dots, \lfloor n\bar{U}_m \rfloor - b_m - 1\}} \frac{T_{nm}(k)}{\sqrt{V_{nm}(k)}}$ 
  if  $\frac{T_{nm'}(k')}{\sqrt{V_{nm'}(k')}} > K_n$  then
    | add  $k'$  to the set of estimated change points
    | WBS-SN( $s, k', K_n, L_0, M, b$ )
    | WBS-SN( $k' + 1, e, K_n, L_0, M, b$ )
  else
    | STOP
  end
end

```

where $T_{nm}^{(r)}(k)$ and $V_{nm}^{(r)}(k)$ are calculated based on the r th sample. Finally, we choose K_n as the $100\gamma\%$ sample quantile of $\{\hat{K}_n^1, \hat{K}_n^2, \dots, \hat{K}_n^R\}$ for some $\gamma \in [0.8, 0.95]$.

Assumption 2. For $i = 1, 2, \dots, m_0$, we assume that

1. $\Delta_i^* = n^{-\delta_i} f_i$ for some $\delta_i \in [0, \frac{1}{2})$ and nonzero element $f_i \in \mathcal{H}$.
2. $K_n \rightarrow \infty$ and $K_n = o(n^{1/2-\bar{\delta}})$ where $\bar{\delta} = \max\{\delta_1, \delta_2, \dots, \delta_{m_0}\}$.

The following theorem shows the estimation consistency for WBS-SN, which is proved in the appendix of the supplement.

Theorem 4. Suppose Assumptions 1 and 2 holds. Let $\hat{m}$ denote the number, and $\hat{k}_1 \leq \hat{k}_2 \leq \dots \leq \hat{k}_{\hat{m}}$ the locations of the change point estimator obtained by the WBS-SN algorithm. Then we have

$$\lim_{n \rightarrow \infty} P(\hat{m} = m_0, |\hat{k}_i - k_i^*| < \epsilon_{in} \text{ for any } i = 1, 2, \dots, m_0) \geq 1 - m_0 \left\{ 1 - \frac{2r(1-2b)}{1-b} \right\}^M, \quad (11)$$

where $\{\epsilon_{in}\}$ is an integer sequence which satisfies that $\frac{\epsilon_{in}}{n} \rightarrow 0$ and $n(\epsilon_{in} \|\Delta_i^*\|_{\mathcal{H}})^{-2} \rightarrow 0$ for $i = 1, 2, \dots, m_0$.

Remark 5. As shown in (11), the asymptotic probability of correctly estimating the number and locations of change points is lower bounded by the same value as for $P(\mathcal{M})$, which converges to one as $M \rightarrow \infty$. Also, the convergence rate ϵ_{in} for the true change point k_i^* depends on the change magnitude $\|\Delta_i^*\|_{\mathcal{H}}$ at k_i^* , so change points of larger magnitude correspond to faster convergence rate. By contrast, a single rate is derived for $\max_{1 \leq i \leq m_0} |\hat{k}_i - k_i^*|$ in Theorem 3 of Zhao, Jiang, and Shao (2022), which develops a SN-based segmentation algorithm for Euclidean valued time series.

4. Simulation Results

In this section, we examine the size and power properties of our proposed test statistic, as well as the estimation accuracy of WBS-SN algorithm, in finite sample. Specifically in Section 4.1, we consider the single change point setting for bivariate distributional data. In Section 4.2, we examine empirical size and power of our test statistic for three data generation processes (DGPs, hereafter) studied in Jiang, Zhu, and Shao (2024b), as well as another DGP where the Fréchet mean and variance remain the same under the alternative. The favorable estimation performance of WBS-SN is shown in Section 4.3. We set $b = 0.15$ for all the simulation results and let $M = 50$, $L_0 = 20$ for the WBS-SN algorithm used in Section 4.3.

4.1. Bivariate Distributional Data

Consider the same DGP as in sec. 3.2 of Dubey and Zheng (2023) and let $\{X_t\}$ take values in the space of bivariate distributions with metric

$$d(x, y) = \left\{ \int_{\mathbb{R}} \int_{\mathbb{R}} |F_x(u, v) - F_y(u, v)|^2 du dv \right\}^{1/2},$$

Table 1. Empirical rejection rate (in percentage) under the null when testing for change point in marginal distribution of bivariate distributional data.

n	ρ	SS-SN	SN ₁	SN ₂	DM	DZ
200	-0.4	4.1	5.2	2.9	6.7	6.0
	0	4.2	5.1	0.9	3.2	6.6
	0.4	6.2	6.7	1.1	25.0	43.4
	0.7	9.2	5.2	4.1	85.1	97.1
400	-0.4	5.1	4.9	3.9	10.7	4.3
	0	4.2	5.5	2.3	4.3	5.5
	0.4	4.3	4.1	0.5	21.2	42.5
	0.7	5.3	5.2	1.4	83.3	95.5

where F_x and F_y are the cdf functions of x and y . The strong negativity of this metric space can be shown in a similar way as (Ω_1, d_W) in Section 2.3. We test for a change point in the marginal distribution of X_t . As comparison, we consider the statistics SN_1 and SN_2 defined in Jiang, Zhu, and Shao (2024b), which have pivotal limiting distributions under the null, and the statistics defined in Dubey and Zheng (2023) (DZ) and Dubey and Müller (2020) (DM) with rejection criteria calculated using 200 bootstrap/permutation replicates. Under the alternative, the size adjusted power for DM and DZ are calculated according to Domínguez and Lobato (2000). The experiments are repeated 1000 times with nominal level at 5%.

For the empirical size, we let $X_t = \mathcal{N}(Z_t, 0.25I_2)$, where I_2 is the two dimensional identity matrix and $Z_t = \rho I_2 Z_{t-1} + 0.5\sqrt{1-\rho^2}\epsilon_t$ with $\epsilon_t \sim \text{iid } \mathcal{N}(0, I_2)$. We set $n \in \{200, 400\}$ and $\rho \in \{-0.4, 0.4, 0, 0.7\}$. As shown in Table 1, our proposed statistic SS-SN has accurate size for $\rho \in \{-0.4, 0, 0.4\}$ and is over-sized for $\rho = 0.7$ when $n = 200$. The empirical sizes are close to the nominal level for all values of ρ when $n = 400$. For DM and DZ, the size distortion is large when $\rho \in \{0.4, 0.7\}$ and the size accuracy does not improve much as n increases. The large size distortion is expected since both DM and DZ are developed for independent object data. SN_2 is under-sized for all values of ρ and the size distortion becomes larger for $\rho \in \{0.4, 0.7\}$ as n increases. For SN_1 , the size accuracy is comparable to SS-SN. However, as will be shown later, SN_1 can only detect change in Fréchet variance.

For the size adjusted power we set $n=200$, $\rho \in \{-0.4, 0, 0.4\}$ and assume $X_t = \mathcal{N}(Z_t, 0.25I_2)$ with Z_t coming from the following two DGPs.

$$\text{DGP1} : Z_t - \mu_t = \rho I_2 (Z_{t-1} - \mu_{t-1}) + 0.5\sqrt{1-\rho^2}\epsilon_t \text{ with } \begin{cases} u_t = (0, 0)^\top, & 1 \leq t \leq \lfloor \frac{n}{2} \rfloor \\ u_t = (\delta, 0)^\top, & \lfloor \frac{n}{2} \rfloor < t \leq n \end{cases}$$

$$\text{DGP2} : Z_t M_t = \rho I_2 Z_{t-1} M_{t-1} + 0.4\sqrt{1-\rho^2}\epsilon_t \text{ with } \begin{cases} M_t = I_2, & 1 \leq t \leq \lfloor \frac{n}{2} \rfloor \\ M_t = \text{diag}(\frac{0.4}{0.4+\delta}, 1), & \lfloor \frac{n}{2} \rfloor < t \leq n \end{cases}$$

Note that for DGP1 only the Fréchet mean of $\{X_t\}$ changes. The size adjusted power against $\delta \in [0, 0.8]$ are plotted in Figure 1 (DM and DZ are not included for $\rho=0.4$ because their size adjusted rejection criteria are both zero, due to the large size distortion under the null). For DGP1, SS-SN achieves the best power as compared to other methods when $\rho = -0.4$ and it has similar performance with the best method in comparison when $\rho \in \{0, 0.4\}$. Except for SN_1 , which has trivial power

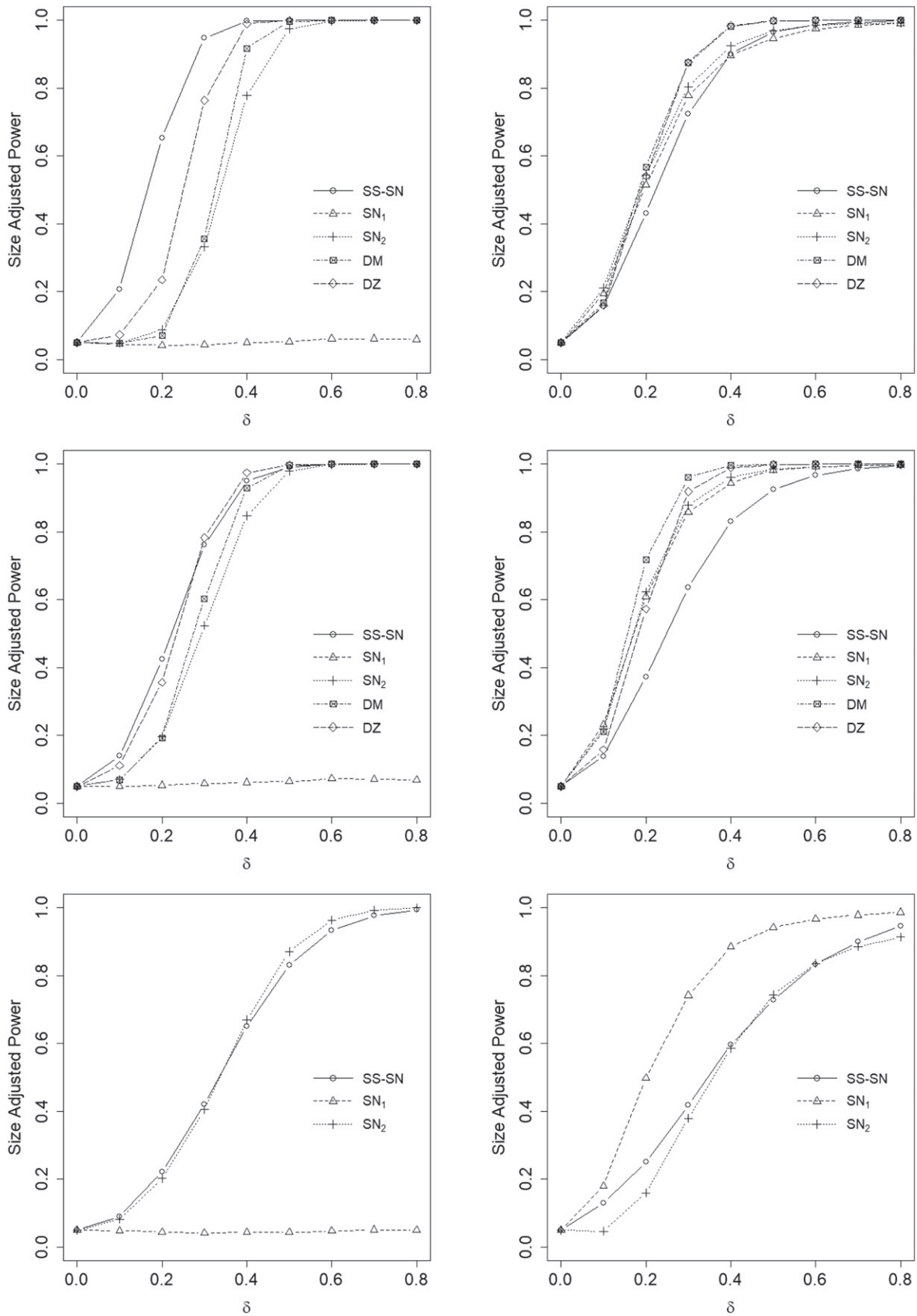


Figure 1. Size adjusted power when testing for change point in marginal distribution of bivariate distributional data from DGP1 (left) and DGP2 (right) with $\rho = -0.4$ (first row), $\rho = 0$ (second row) and $\rho = 0.4$ (last row).

Table 2. Empirical rejection rate (in percentage) under the null for DGP3-6.

n	ρ	DGP3			DGP4			DGP5			DGP6		
		SS-SN	SN_1	SN_2	SS-SN	SN_1	SN_2	SS-SN	SN_1	SN_2	SS-SN	SN_1	SN_2
200	-0.4	3.9	4.1	3.4	4.0	4.3	3.6	4.7	5.3	2.1	3.7	5.4	3.6
	0	4.1	5.0	1.5	3.5	6.1	3.2	6.0	4.5	2.9	4.6	6.0	2.8
	0.4	5.1	6.3	3.7	6.3	5.0	4.1	6.5	4.6	3.5	6.5	6.8	1.8
	0.7	7.3	7.6	10.6	8.5	4.2	10.5	7.4	4.5	5.3	9.7	6.9	0.2
400	-0.4	5.0	6.2	5.3	4.8	5.1	3.8	5.0	5.4	2.2	4.5	6.3	6.0
	0	4.1	6.2	3.3	4.6	6.0	2.5	5.5	5.6	2.4	7.0	5.7	4.8
	0.4	5.4	6.5	2.6	4.7	6.4	2.6	4.6	5.4	2.1	7.1	6.4	2.0
	0.7	5.7	6.5	6.3	4.7	5.2	7.6	5.6	4.7	3.0	8.0	7.1	0.6
800	-0.4	7.1	5.6	4.9	4.6	5.5	4.7	4.1	5.0	2.5	6.0	5.4	5.2
	0	5.7	5.3	3.2	5.1	5.0	3.2	4.2	5.2	2.6	5.3	5.1	4.4
	0.4	6.3	4.5	2.2	5.5	4.2	1.6	5.1	4.7	1.7	5.6	5.8	3.4
	0.7	7.2	4.8	3.4	5.4	5.6	4.0	4.8	5.2	2.0	6.1	5.5	1.2

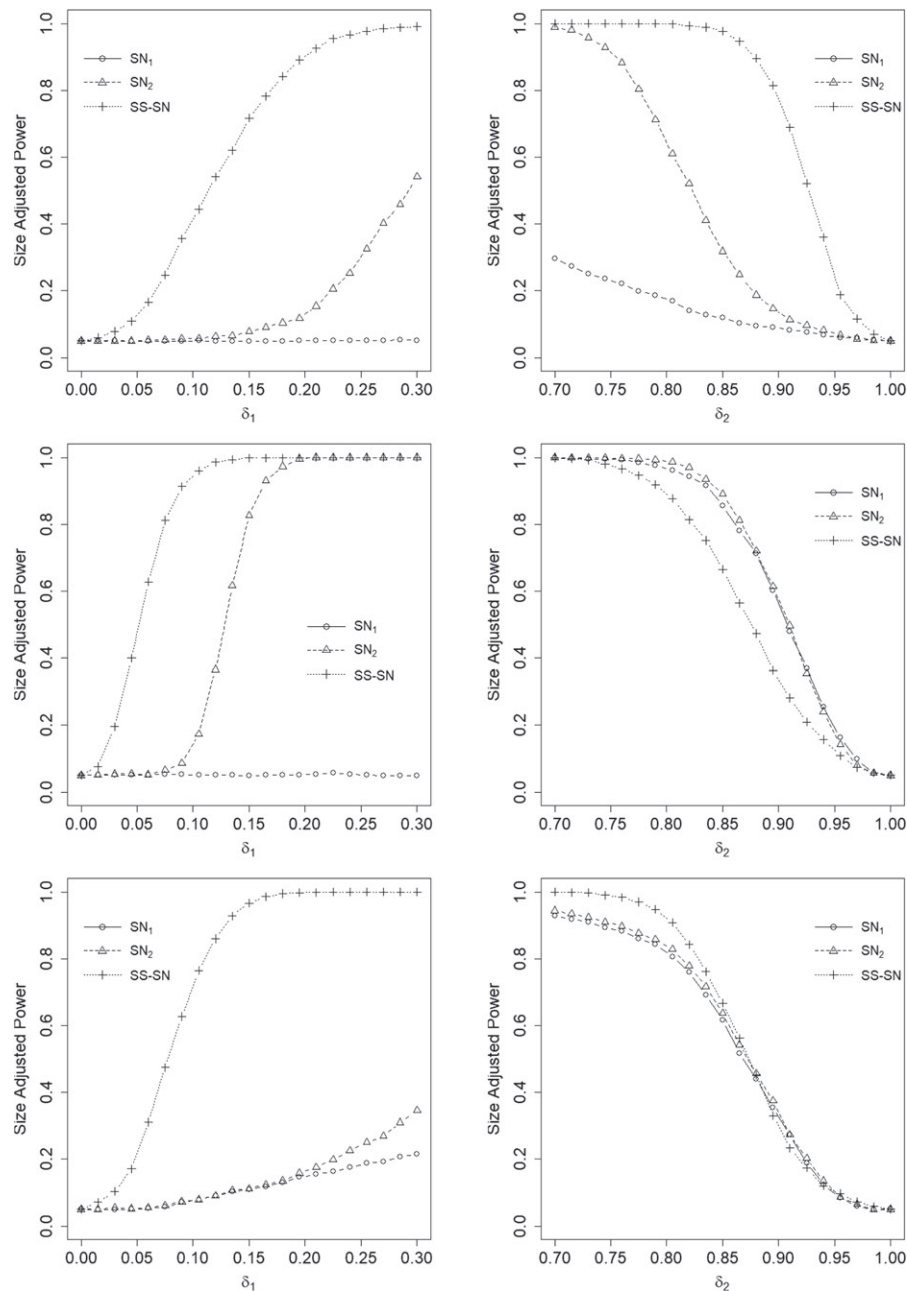
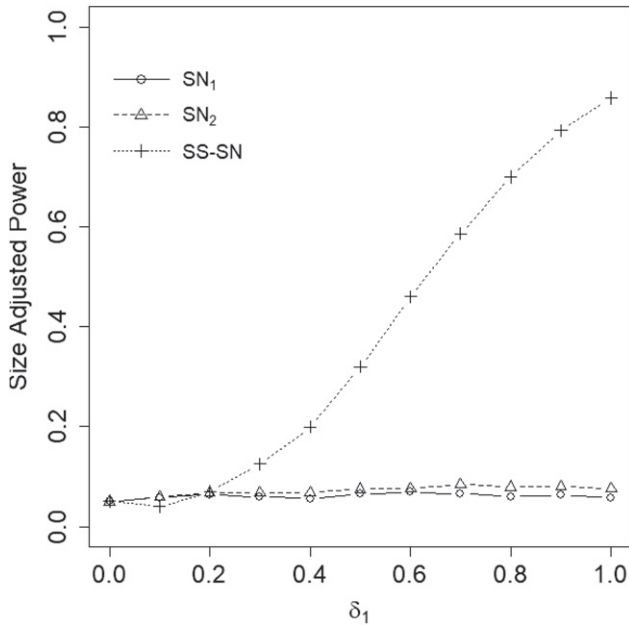
**Figure 2.** Size adjusted power for testing hypothesis on the existence of a change point for ALT1 (first column) and ALT2 (second column) and for DGP4 (first row), DGP5 (second row) and DGP6 (last row).

Table 3. Simulation results for multiple change point detection problems based on 200 Monte Carlo repetitions.

ρ	Case	Method	Proportions (%)				ARI
			0	1	2	≥ 3	
-0.5	(i)	Ours	0	13.78	84.69	1.53	0.889
		Jiang et al.	0.51	45.92	51.02	2.55	0.734
	(ii)	Ours	0	2.59	95.85	1.55	0.956
		Jiang et al.	0	5.70	90.16	4.15	0.915
0	(i)	Ours	0	25.64	74.36	0	0.835
		Jiang et al.	0.51	37.43	61.54	0.51	0.772
	(ii)	Ours	0	0	99.5	0.5	0.971
		Jiang et al.	0	4.5	91.5	4	0.936
0.5	(i)	Ours	5	48.5	43.5	3	0.639
		Jiang et al.	9	46.5	36	8.5	0.589
	(ii)	Ours	0	8.67	81.12	10.20	0.907
		Jiang et al.	2.55	16.33	62.76	18.37	0.796
0.7	(i)	Ours	9	43	34	14	0.549
		Jiang et al.	10.5	36	32	21.5	0.525
	(ii)	Ours	1.51	15.08	57.29	26.13	0.800
		Jiang et al.	3.02	23.62	43.22	30.15	0.663

NOTE: Proportions for detected number of change points and adjust Rand index (ARI) are reported.

**Figure 3.** Size adjusted power for testing hypothesis on the existence of a change point for DGP3.

under DGP1, SN_2 has the largest power loss when $\rho \in \{-0.4, 0\}$. For DGP2, SS-SN has mild power loss when $\rho = -0.4$ and the power loss becomes larger as ρ increases, while it slightly outperforms SN_2 when $\rho = 0.4$.

4.2. Univariate Distribution, Graph Laplacian and Covariance Matrix

In this section, we consider three DGPs of object valued time series studied in Jiang, Zhu, and Shao (2024b), as well as another DGP where the Fréchet mean and variance remain the same under the alternative. The experiments are repeated 1000 times with nominal level at 5%. For DGP3 and DGP4, $\{X_t\}$ takes value in the space of probability measures on $\mathbb{R}$ equipped with the 2-Wasserstein metric d_W . For DGP5, $\{X_t\}$ resides in the space of Graph Laplacian of weighted graphs equipped with Frobenius

metric d_F and for DGP6, $\{X_t\}$ takes value in the space of positive definite covariance matrices with the log-Euclidean metric d_E .

Let $U_t = (U_{t,1}, U_{t,2})$ be generated from the VAR(1) process $U_t = \rho I_2 U_{t-1} + \epsilon_t$, where $\{\epsilon_t\}$ are iid standard bivariate normal random vectors, and define $\beta_t \sim_{\text{iid}} \text{Bernoulli}(\delta_1)$, $e_t \sim_{\text{iid}} \text{EXP}(1)$ such that $\{\epsilon_t\}$, $\{\beta_t\}$ and $\{e_t\}$ are independent. Assume $X_t = X_t^{(1)}$ for $1 \leq t \leq \lfloor n/2 \rfloor$ and $X_t = X_t^{(2)}$ for $1 + \lfloor n/2 \rfloor \leq t \leq n$, we consider the following four DGPs. DGP3: $X_t^{(1)} = \mathcal{N}(U_{t,1} + 1, 1)$ and $X_t^{(2)} = \mathcal{N}((1 - \beta_t)(U_{t,1} + 1) + \beta_t e_t, 1)$. DGP4: $X_t^{(1)} = \mathcal{N}(\arctan(U_{t,1}), [\arctan(U_{t,1}^2) + 1]^2)$ and $X_t^{(2)} = \mathcal{N}(\arctan(U_{t,1}) + \delta_1, \delta_2^2 [\arctan(U_{t,1}^2) + 1]^2)$. DGP5: Graph Laplacians where each graph has five nodes that are categorized into two communities with two and three nodes, respectively, and the edge weights for the first community, the second community and between communities are set as $0.4 + \arctan(U_{t,1}^2)$, $0.2 + \arctan(U_{t,2}^2)$, 0.1 for $X_t^{(1)}$ and $\delta_2[0.4 + \arctan(U_{t,1}^2)]$, $\delta_2[0.2 + \arctan(U_{t,2}^2)]$, $0.1 + \delta_1$ for $X_t^{(2)}$. DGP6: Covariance matrix $X_t^{(i)} = (2I_3 + Z_{t,i})(2I_3 + Z_{t,i})^\top$ for $i = 1, 2$, where all the entries of $Z_{t,1}$ (resp. $Z_{t,2}$) are independent copies of $\arctan(U_{t,1})$ (resp. $\delta_1 + \delta_2 \arctan(U_{t,1})$). Since the applicability of both DM and DZ is limited to independent object data, we shall not include them for comparison as they are expected to exhibit large size distortion when temporal dependence is present as in Section 4.1; also see Jiang, Zhu, and Shao (2024b).

For the empirical size, we set $\delta_1 = 0$, $\delta_2 = 1$, $n \in \{200, 400, 800\}$ and $\rho \in \{-0.4, 0.4, 0, 0.7\}$. As shown in Table 2, SS-SN is mildly over-sized when $\rho = 0.7$ for DGP3 and DGP6, as well as when $n = 200$, $\rho = 0.7$ for DGP4 and DGP5. The size accuracy for SN_1 and SS-SN is comparable for $\rho \in \{-0.4, 0, 0.4\}$, while SN_1 has more accurate size when $\rho = 0.7$. Similar to the empirical size result in Section 4.1, SN_2 is under-sized for most combinations of (n, ρ) , except when $n = 200$ and $\rho = 0.7$ for DGP3 and DGP4, in which case SN_2 is over-sized.

For the size adjusted power of DGP4, DGP5, and DGP6, we fix $n = 400$, $\rho = -0.4$ and consider two alternatives. ALT1: Fix $\delta_2 = 1$ and plot the size adjusted power against $\delta_1 \in [0, 0.3]$. ALT2: Fix $\delta_1 = 0$ and plot the size adjusted power against $\delta_2 \in [0.7, 1]$. For DGP4 and DGP5 (see sec. 5 Jiang, Zhu,

and Shao 2024b), the Fréchet mean changes and the Fréchet variance remains the same under ALT1, while ALT2 has a break in Fréchet variance with Fréchet mean being held fixed. For DGP6, both Fréchet mean and variance changes under ALT1 or ALT2. As shown in Figure 2, SS-SN outperforms SN_1 and SN_2 in all settings except DGP2 under ALT2. Compared with SS-SN, the power loss for SN_1 and SN_2 are very large under ALT1 (as in Section 4.1, SN_2 has trivial power under ALT1 for DGP4 and DGP5) and for DGP4 under ALT2, while all three methods have similar performance under ALT2 for DGP6.

For the size adjusted power of DGP3, we fix $n = 800$, $\rho = -0.4$ and the size adjusted power against $\delta_1 \in [0, 1]$ is plotted in Figure 3. Note that the Fréchet mean and variance are fixed at $\mathcal{N}(1, 1)$ and 1 before and after the change. For this DGP, SN_1 and SN_2 both have trivial power while SS-SN successfully detected the change in marginal distribution.

4.3. Multiple Change Point Estimation

For the setting with multiple change points, we compare our method with Jiang, Zhu, and Shao (2024b), which combines WBS and SN_2 test statistic proposed therein for the object-valued time series. The sample size is set to be $n = 200$ to be comparable with the sample size in applications. Similar to the simulations conducted in Jiang, Zhu, and Shao (2024b), we employ Wasserstein distance and consider the following data generating process: for $t = 1, 2, \dots, 200$,

$$X_t = \mathcal{N}(\arctan(U_t) + \delta_{t,1}, \delta_{t,2}^2[\arctan(U_t^2) + 1]^2),$$

where $\{U_t\}$ is a AR(1) process, that is, $U_t = \rho U_{t-1} + \epsilon_t$, $\epsilon_t \stackrel{iid}{\sim} \mathcal{N}(0, 1)$; $\{\delta_{t,1}\}$ and $\{\delta_{t,2}\}$ reflecting change points locations are defined as

$$\delta_{t,1} = a_1 \mathbf{1}_{\{t \leq 65\}} + a_2 \mathbf{1}_{\{65 < t \leq 135\}} + a_3 \mathbf{1}_{\{135 < t \leq 200\}},$$

$$\delta_{t,2} = b_1 \mathbf{1}_{\{t \leq 65\}} + b_2 \mathbf{1}_{\{65 < t \leq 135\}} + b_3 \mathbf{1}_{\{135 < t \leq 200\}}.$$

Constants a_i and b_i determine the signal strength and we consider the following two cases: (i) $(a_1, a_2, a_3) = (0, 0.7, 0.1)$ and $(b_1, b_2, b_3) = (1.2, 1.5, 1.8)$; (ii) $(a_1, a_2, a_3) = (-0.1, 0, 0.1)$

and $(b_1, b_2, b_3) = (1, 1.6, 0.9)$. The parameter ρ determines the level of dependency between data points and its effects on the change point detection are investigated in this simulation. Table 3 reports the proportions for detected number of change points and adjusted Rand index (ARI) based on 200 Monte Carlo repetitions. Note that $ARI \in [0, 1]$ measures the accuracy of change point estimation and larger ARI corresponds to more accurate estimation. We can observe that our method is uniformly better than Jiang, Zhu, and Shao (2024b) as reflected by ARI. As the performance declines for both methods with increasing values of ρ , the method proposed by Jiang, Zhu, and Shao (2024b) tends to more severely overestimate the number of change points.

5. Applications

In this section, we apply our WBS-SN algorithm to datasets arising from the stock market and the taxi traffic in New York city. We set $b = 0.15$, $M = 50$ and $L_0 = 20$ for the WBS-SN algorithm.

5.1. Stock Return Data

Density function for log-returns of S&P 500 companies can reflect the overall stock market situation. In this study, we are interested in monitoring abrupt changes in marginal distributions of the time series of log-return densities. The daily log-returns of a stock can be downloaded using R package “BatchGetSymbols” (Perlin 2022). For each month from year 2015 to year 2022, a density function is constructed from the daily log-returns of companies listed in S&P 500, resulting in a total of 96 density functions. These densities are constructed using a kernel density estimator with the bandwidth set to be one-tenth of range between 1st and 99th percentiles of all log-returns. The heat map is shown in Figure 4 with each column representing a density function evaluated on 51 grid points. Our test indicates significant changes in the marginal distributions of the monthly density time series before and after January 2020,

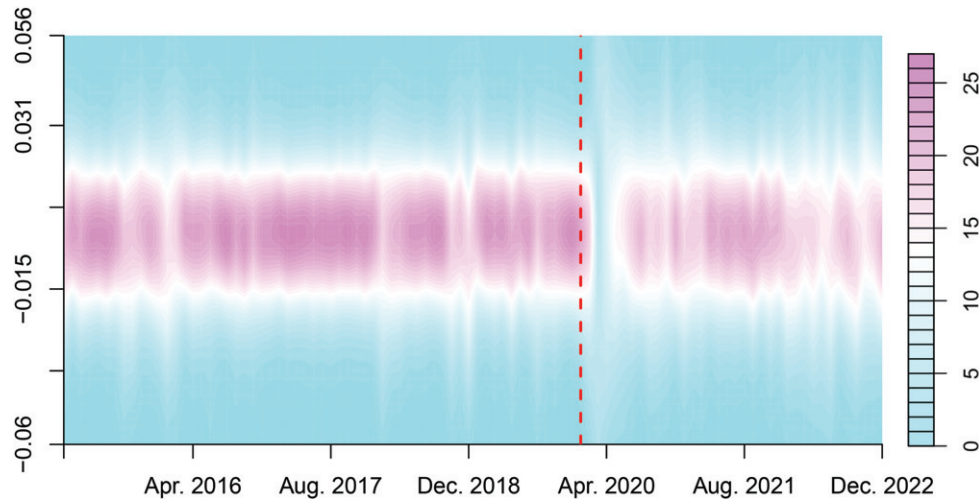


Figure 4. Heat plot of 96 density functions. Each column represents a density function. The vertical dashed red line represents the detected change point location by our method.

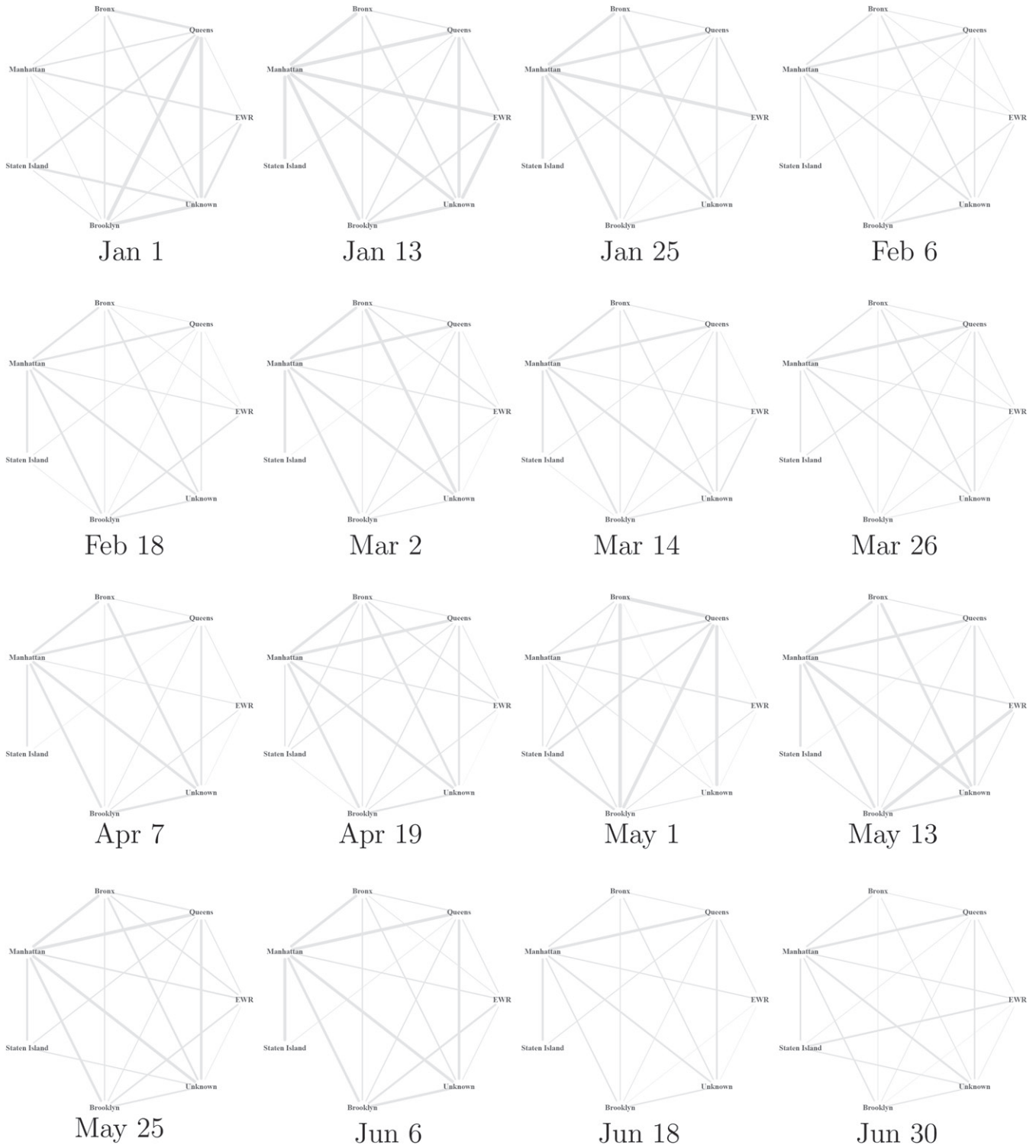


Figure 5. Visualization of weighted undirected taxi-trip networks indexed by date.

aligning with the onset of 2020 stock market downturn and increasing volatility due to the COVID-19 pandemic.

5.2. TLC Trip Record Data

In this section, we apply our WBS-SN algorithm to datasets arising from the taxi traffic in New York city. To be specific, we aim to analyze the taxi traffic patterns in New York city

through a change point perspective. The taxi traffic situations can be summarized using daily networks. Using the TCL trip record data (<https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page>), for each day between January 2023 and June 2023, we construct a weighted undirected network comprising seven nodes. Six of these nodes represent Newark Airport (EWR), Queens, Bronx, Manhattan, Staten Island, Brooklyn in New York, while the seventh node represents areas outside these regions. For any two nodes $i, j \in \{$

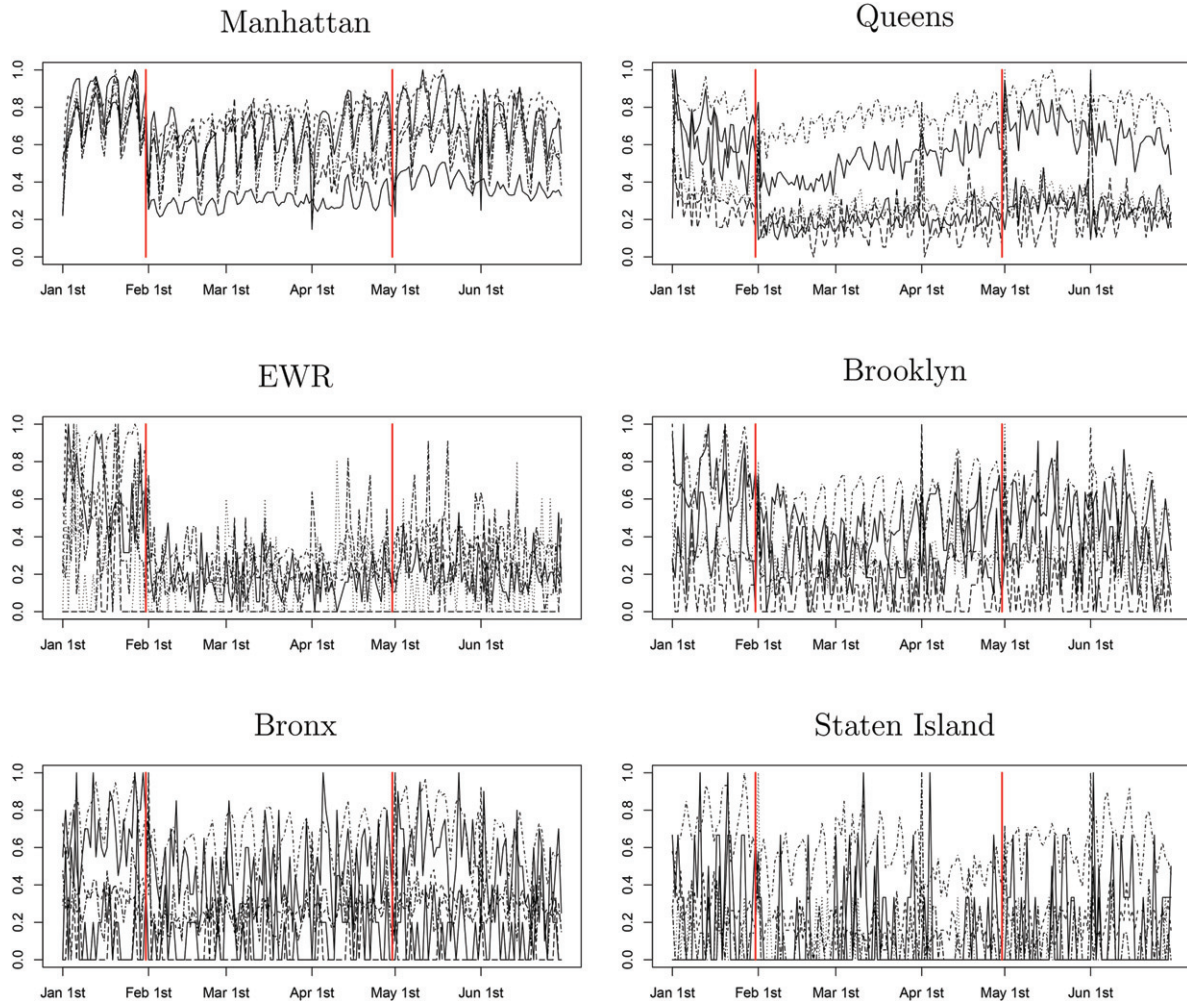


Figure 6. For each node $i \in \{\text{Manhattan, Queens, EWR, Brooklyn, Bronx, Staten Island}\}$, the edge weights $\{w_{ij,t} : j \neq i\}$ are plotted as multivariate time series.

EWR, Queens, Bronx, Manhattan, Staten Island, Brooklyn, Unknown }, let $m_{ij,t}$ represents the number of taxi trips originating in one of the two nodes and ending in the other on day t . The weight $w_{ij,t}$ on the edge of the network on day t between node i and node j is set to be $w_{ij,t} = m_{ij,t} / \max_t\{m_{ij,t}\}$ if $\max_t\{m_{ij,t}\} > 0$. Self-edges, that is, trips that start and end in the same region, are excluded from our analysis. This yields a network time series spanning 181 days. A selection of these daily networks are shown in Figure 5 with the edge width corresponding to the magnitude of $w_{ij,t}$. Two change points, located on January 31 and April 30, are identified by our method, indicating significant differences in the marginal distributions of the network time series before and after these dates. To better interpret the result, we plot the edge weights $\{w_{ij,t} : j \neq i\}$ as multivariate time series for each region in Figure 6. For Manhattan, a shift in mean of the edge weights are clearly observed before and after each of the change point locations.

6. Conclusion

In this article, we propose a novel change point detection method for a broad class of object-valued time series that can

detect marginal distributional changes. By using Hilbert space embedding, sample splitting and self-normalization, the statistic we developed is almost tuning parameter free, has pivotal limiting null distribution and only uses the distance between two non-Euclidean data objects. For multiple change point estimation, the estimation consistency and convergence rate of change point estimators are established for the proposed WBS-SN algorithm, and the convergence rates for each change point estimator are adaptive to the heterogeneous change magnitudes. To the best of our knowledge, we are the first to show the WBS estimation consistency for a broad class of object-valued time series and in a nonparametric setting. Extensive simulations are conducted to show the accurate size and satisfactory power performance of the proposed method in comparison with existing ones. The data examples further illustrate the versatility of our proposed method in real-world applications.

To conclude, we mention several possible extensions. For multiple change point estimation, this article only considers unknown but fixed number of change points. It would be helpful to consider the setting where the number of change points can grow with sample size. Also, it may be desirable to relax the abrupt change assumption and develop a test that captures gradual change.

Supplementary Materials

The online supplementary material contains the proofs for Theorems 2, 3 and 4.

Disclosure Statement

The authors report there are no competing interests to declare.

Data Availability Statement

All data used in this research are publicly available. The stock return data can be freely downloaded from Yahoo Finance (<https://finance.yahoo.com/>) and the TCL trip record data is available at <https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page>.

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